Week 50: Staircases and lengths

Adult guide built from the problem files · Unpiloted. Physical pretests unperformed.

This week's adult guide is week-50-facilitator.pdf, whose source is editable/facilitator-src/content.json with route-note.json. These problem files hold each problem's one-sentence idea, not its answers; the answers, hints and the turn proof are in the guide.

Problems and routes

Problem	K–1	2–3	4–5	route
Two different staircases	P1	P1	P1	first
A one-step and a two-step staircase	P2	P2	P2	first
A staircase closer to the diagonal	P3	P3	P3	first
Fewest turns inside the strip	P4	P4	P4	return
Paths that may go left	P5	P5		return
A leftward piece and less string	P6			return
A narrower strip			P5	return
The exact length of a path that goes left once			P6	return
Could more, smaller pieces reach 200 mm?			P7	return

First route. K–1 P1–P3; Grades 2–3 P1–P3; Grades 4–5 P1–P3. **Return visits.** K–1 P4–P6; Grades 2–3 P4–P5; Grades 4–5 P4–P7.

Key, one entry per problem

Two different staircases

K–1 P1 · Grades 2–3 P1 · Grades 4–5 P1 · first route

Idea. Every right-and-up path from S to F has length width + height = 280 mm.

A one-step and a two-step staircase

K–1 P2 · Grades 2–3 P2 · Grades 4–5 P2 · first route · wording differs by band

Idea. Regrouping the right pieces and the up pieces shows the two lengths are equal.

A staircase closer to the diagonal

K–1 P3 · Grades 2–3 P3 · Grades 4–5 P3 · first route · wording differs by band

Idea. Staircases can come as close to the diagonal as you like, yet every one stays 280 mm long; the diagonal is 200 mm.

Fewest turns inside the strip

K–1 P4 · Grades 2–3 P4 · Grades 4–5 P4 · return route · wording differs by band

Idea. Inside the 15 mm strip the fewest turns is 8: a horizontal piece spans at most 40 mm, and 20 mm at either end.

Paths that may go left

K–1 P5 · Grades 2–3 P5 · return route · wording differs by band

Idea. With total leftward travel L the length is 280 + 2L mm.

A leftward piece and less string

K–1 P6 · return route

Idea. Every millimetre moved left must be made up by an extra millimetre right, so a leftward piece adds string.

A narrower strip

Grades 4–5 P5 · return route

Idea. Staying closer to the diagonal does not shorten a staircase: one inside an 8 mm strip is still 280 mm.

The exact length of a path that goes left once

Grades 4–5 P6 · return route

Idea. The right and left pieces still end 160 mm to the right, but together they travel more than 160 mm.

Could more, smaller pieces reach 200 mm?

Grades 4–5 P7 · return route

Idea. Staircases approach the diagonal in position but not in length.